

Visual-Inertial ESEKF SLAM

from First Principles

A verified stereo-inertial pipeline, its real-data behavior, and an honest consistency diagnosis

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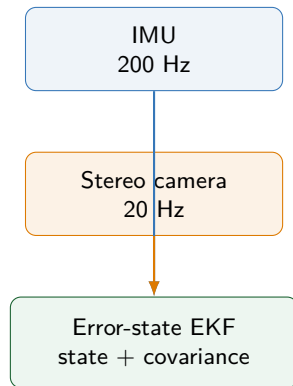
Technical deep dive

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The question: can a small estimator be trustworthy enough to teach from?

Build a visual-inertial SLAM system from scratch, then make every important claim testable.

- Stereo camera: geometric constraints, but intermittent and data-association dependent.
- IMU: high-rate motion prediction, but bias turns into rapid position drift.
- Target: an offline EuRoC pipeline that can become a real-time ROS 2 system.



The hard part is not integrating motion; it is managing uncertainty

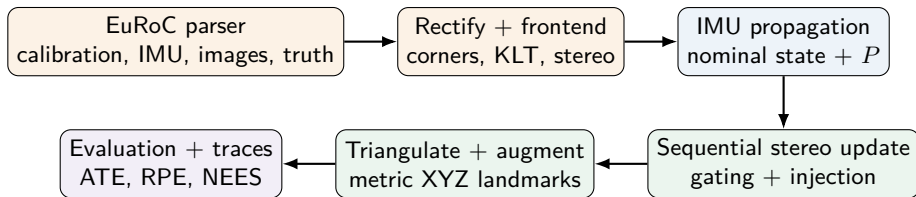
- Accelerometer bias is integrated twice into position.
- Gyroscope bias rotates gravity into a false horizontal acceleration.
- Vision can correct drift only if projection, frames, Jacobians, and covariance all agree.

A plausible trajectory is not evidence that the covariance is correct.

Design goal

Every layer gets an independent test oracle: analytic truth, finite differences, dense batch update, or Monte Carlo.

The project is an end-to-end visual-inertial system, not only an EKF equation



- Implemented: parser, rectification, KLT frontend, propagation, stereo update, augmentation, full MH_01 evaluator.
- Not implemented: marginalization, FEJ/OC-EKF, iterated EKF, ROS 2 deployment.

The nominal state lives on $SO(3)$; the filter estimates a local error

$$x = [p_W, v_W, R_{WB}, b_a, b_g]$$

$$\delta x = [\delta p_W, \delta v_W, \delta \theta_B, \delta b_a, \delta b_g] \in \mathbb{R}^{15}$$

- Position, velocity, and biases are additive.
- Rotation is corrected multiplicatively:

$$R_{WB} \leftarrow R_{WB} \exp(\delta \theta_B^\wedge).$$

Why error-state?

A 3D local rotation error has an ordinary Gaussian covariance; the nominal rotation remains a valid element of $SO(3)$ after every update.

Convention is load-bearing

$\delta \theta$ is body-frame, right-multiplied. Mixing left and right perturbations can pass trivial tests and still corrupt a real filter.

Propagation uses IMU measurements to advance state and covariance

$$a_W = R_{WB}(a_m - b_a - n_a) + g_W$$

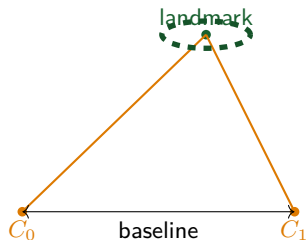
$$p^+ = p + v\Delta t + \frac{1}{2}a_W\Delta t^2, \quad v^+ = v + a_W\Delta t$$

$$R^+ = R \exp((\omega_m - b_g - n_g)\Delta t)^\wedge$$

$$P^+ = \Phi P \Phi^T + Q_d, \quad \Phi \approx I + F\Delta t$$

- Nominal state is integrated on the manifold.
- Covariance is propagated in the 15D local error coordinates.
- First-order Φ is a deliberate, measured trade-off.

Stereo converts matched pixels into a metric geometric constraint



- Rectification makes correspondence a row-constrained search.
- Disparity gives metric depth, but depth uncertainty grows along the viewing ray.
- New landmarks inherit uncertainty from pixels *and* robot pose.

A camera update is a pixel-space correction, not a direct position measurement

For a landmark p_L^W , project it into both rectified cameras:

$$y = [u_0 \quad v_0 \quad u_1 \quad v_1]^T, \quad \nu = y - h(\hat{x})$$

$$S = HPH^T + R, \quad K = PH^T S^{-1}$$

$$\delta x = K\nu, \quad P \leftarrow (I - KH)P(I - KH)^T + KRK^T$$

Interpretation

The innovation asks: “How surprising is this pixel observation under the current state and covariance?”

Gate

Reject an observation when $\nu^T S^{-1} \nu$ exceeds the 4-dof chi-square threshold.

Sequential updates preserve a useful batch-update oracle

- A frame can contain many landmarks, but each stereo observation has only nine non-zero state columns.
- The implementation freezes the nominal linearization point at frame entry, sweeps landmarks sequentially, and injects one accumulated error at the end.
- That design makes the sequential result equal the dense batch update for the same frame; a test verifies agreement to 10^{-12} .

Why this matters in an interview

The batch path is not production code. It is a high-cost oracle that narrows an inconsistency from “somewhere in the filter” to a specific component.

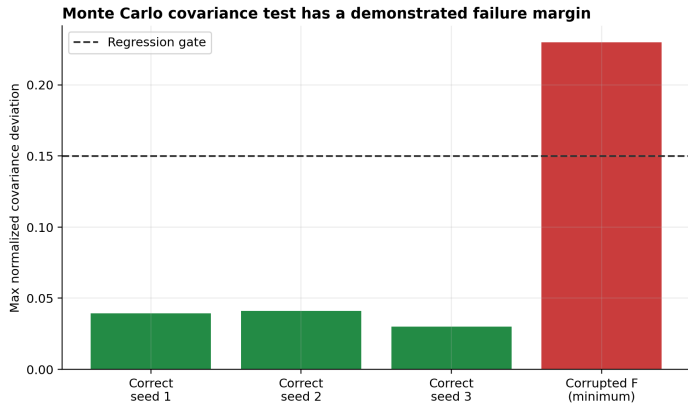
The test strategy targets failure modes, not just coverage

Evidence	What it can falsify
Analytic synthetic truth	Integrator order, gravity sign, bias removal
Central differences	Frame/sign/order mistakes in projection Jacobians
Dense batch oracle	Sequential sweep or accumulated-error mistakes
Monte Carlo covariance	Transposed, sign-flipped, or mis-scaled F/Q blocks
EuRoC MH_01	Parser, calibration, frontend, timing, and closed-loop behavior

The

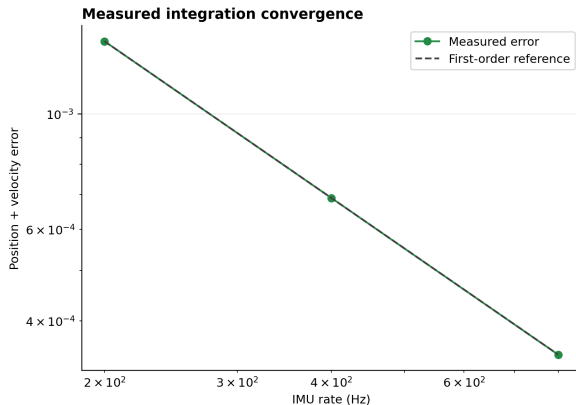
strongest tests compare independently produced behavior, not two versions of the same algebra.

Monte Carlo makes covariance claims falsifiable



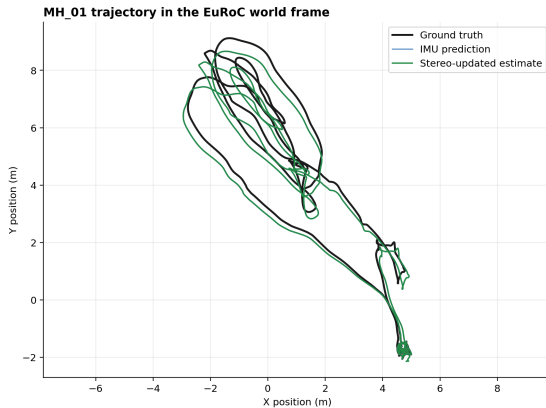
Correct propagation: three seeded runs. Corrupted- F bar: minimum observed deviation over single-block corruptions.

The measured integrator order matches the chosen discretization



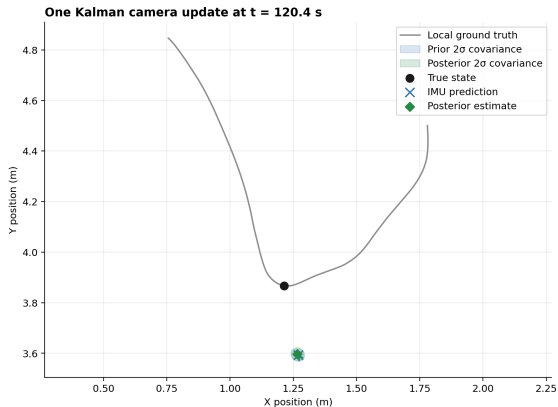
- Halving Δt halves the combined position/velocity error.
- The second-order position term improves the constant, not the global order of the method.

MH_01 runs end-to-end in the raw EuRoC world frame



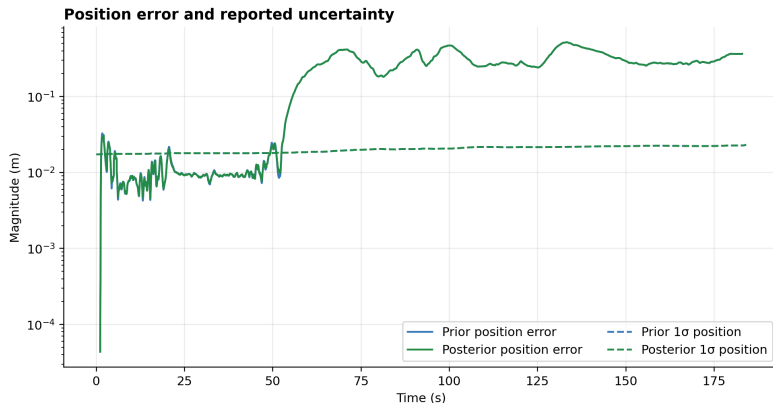
Truth-initialized full MH_01 run. No global SE(3) alignment: alignment would hide drift.

A Kalman update has a prior, a posterior, and uncertainty geometry



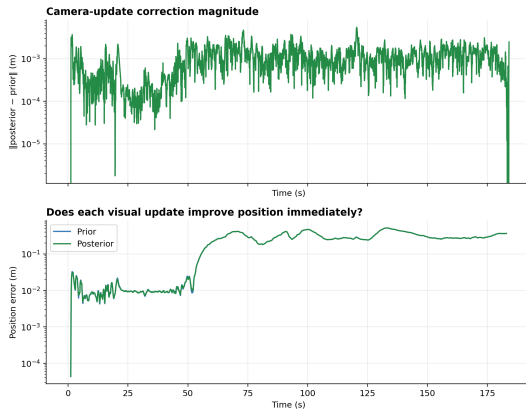
The figure shows state-space prior/posterior and 2-sigma position covariance. The actual camera observation remains in pixel space.

The real run exposes over-confidence rather than hiding it



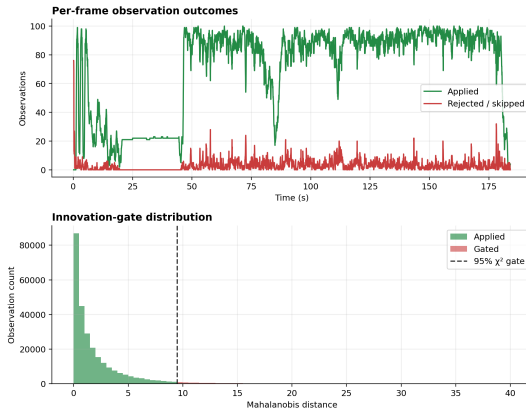
- After roughly 50 seconds, position error is much larger than reported 1-sigma position uncertainty.
- This is an observation from one real trajectory; the controlled Monte Carlo experiment supports the statistical consistency claim.

Camera updates are visible as small, frequent posterior corrections



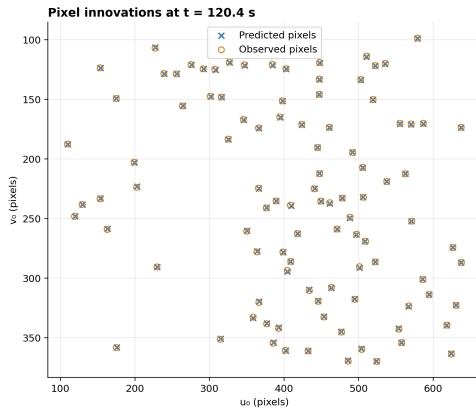
Prior is the IMU-propagated camera-time state; posterior is after the visual sweep.

Gating reveals the estimator's view of innovation consistency



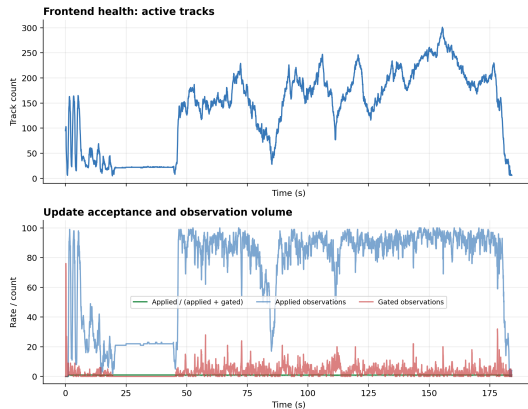
- Applied observations lie predominantly below the 4-dof chi-square gate.
- Gating is a covariance-aware consistency check, not a replacement for frontend outlier rejection.

Pixel innovations make the measurement model concrete



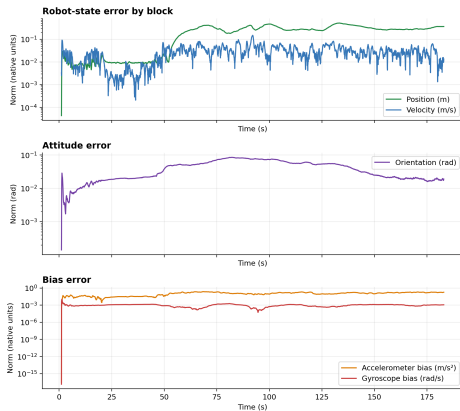
Each gray segment is a predicted-to-observed left-camera pixel innovation for one accepted feature at one camera time.

Frontend health connects visual tracking to estimator updates



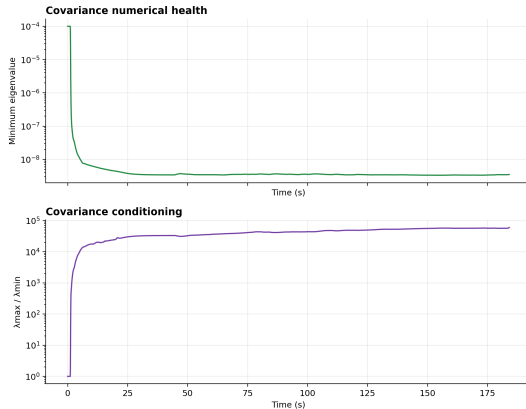
- Track count is not accuracy, but sudden loss constrains how much visual information the filter can receive.
- The acceptance fraction contextualizes the update count with the gate's decision.

The full trace separates position, velocity, attitude, and bias errors



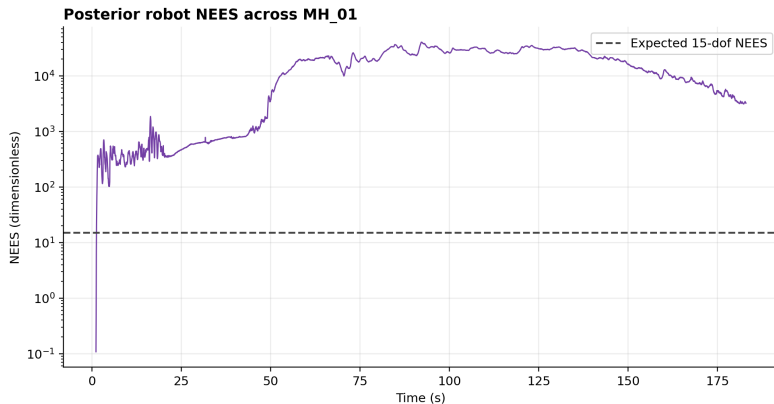
Native units differ by block; compare within a panel, not across panels.

The covariance stays positive but becomes increasingly ill-conditioned



- Positive minimum eigenvalue is a necessary numerical-health condition.
- Conditioning is a monitoring signal; it is not itself proof of consistency.

NEES turns “looks reasonable” into a consistency question



Read this correctly

A pointwise real-run NEES trace identifies over-confidence. A chi-square confidence interval applies to an ensemble mean, which is why the synthetic Monte Carlo control is essential.

The closed-loop synthetic test improves drift but fails consistency

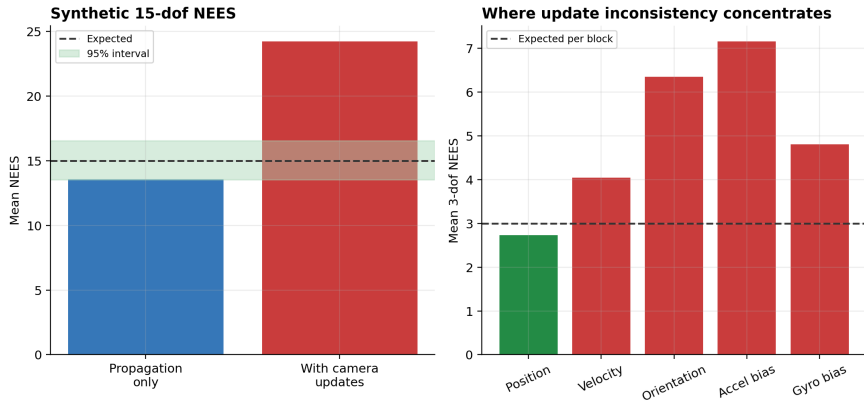
Same 2-second scenario

- Propagation-only final position error: **0.163 m**
- With camera updates: **0.0545 m**
- 195 applied observations; 5 gated

But mean 15-dof NEES

	Mean	95% interval
Propagation only	13.56	[13.52, 16.56]
With updates	24.23	[13.52, 16.56]

The failure is concentrated in attitude and bias, not position



Synthetic 50-run Monte Carlo diagnostic. The left panel uses the 15-dof mean interval; the right uses expected 3 per 3-dof block.

Evidence rules out the most obvious explanations

- ❶ **Not propagation covariance:** the propagation-only control passes in the same harness.
- ❷ **Not sequential update mechanics:** sequential and dense batch agree to 10^{-12} .
- ❸ **Not Jacobian convention algebra:** measurement Jacobians agree with central differences to 10^{-5} .
- ❹ **Not gate selection bias:** widening the gate to infinity changes total NEES only $24.23 \rightarrow 23.66$.
- ❺ **Not yaw observability:** yaw NEES is 1.41 for expected 1.0; tilt NEES is 4.81 for expected 2.0.

The diagnosis is first-order rotation linearization, not FEJ

- Reducing initial tilt sigma from 0.01 rad to 0.001 rad moves total NEES from 24.23 to 15.92, inside the target interval.
- Tilt enters the camera model nonlinearly through rotation; position and landmark position are linear when R is fixed.
- An iterated EKF relinearizes toward the posterior, targeting this measured second-order error.

Engineering decision

Do not tune the consistency bound until it passes. Keep a regression ceiling and retain the statistical acceptance test as disabled future-work evidence.

What this project demonstrates

- ① **Estimator fluency:** manifold error states, stochastic IMU propagation, stereo projection, gated EKF updates, and landmark augmentation.
- ② **Systems fluency:** real image I/O, calibration/rectification, feature tracking, timing, data association, and reproducible traces.
- ③ **Scientific discipline:** tests designed to fail under plausible mistakes, plus an explicit measurement of an unresolved limitation.

Bottom line

A working trajectory is the start of estimator validation, not the end. The deliverable is a system whose errors are measurable, explainable, and actionable.

The next iteration is targeted by evidence

Estimator

- Iterated EKF for tilt-linearization error.
- Static-start gravity alignment.
- Marginalization / bounded-map strategy.

System

- Calibrated frontend noise and stronger outlier rejection.
- Multi-sequence EuRoC evaluation.
- ROS 2 + Jetson timing validation.

Thank you

Questions

Appendix: map of the deeper technical material

- Frame and perturbation conventions
- Transition-matrix block structure
- Update sequence and reset Jacobian
- Stereo triangulation uncertainty
- Full MH_01 trace schema
- Metrics and their limitations
- Runtime/profile findings
- Interview questions and direct answers

Appendix: frame convention and perturbation rule

Quantity	Convention
p_W, v_W	Position and velocity expressed in world W
R_{WB}	Rotates body-frame vectors into world frame
$\delta\theta_B$	Body-frame local orientation error
T_{BS}	Sensor-to-body rigid transform
P ordering	$[p, v, \theta, b_a, b_g]$

$$R_{WB}^{\text{true}} = R_{WB}^{\text{estimate}} \exp(\delta\theta_B^\wedge)$$

The finite-difference test intentionally rejects a left/global perturbation, preventing a convention mismatch from becoming a latent real-data bug.

Appendix: the error transition matrix exposes the critical couplings

$$F = \begin{bmatrix} 0 & I & 0 & 0 & 0 \\ 0 & 0 & -R[a - b_a]_{\times} & -R & 0 \\ 0 & 0 & -[\omega - b_g]_{\times} & 0 & -I \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

- $F_{v\theta}$ couples an orientation error into a wrong world acceleration.
- F_{vb_a} and $F_{\theta b_g}$ carry accelerometer/gyro bias uncertainty into navigation error.
- These blocks are exactly why identity-state, hand-calculated tests are insufficient.

Appendix: sequential update mechanics

- 1 Freeze nominal state at frame entry.
- 2 For each known, visible landmark: project, build sparse H_i , compute S_i , and gate against the prior covariance.
- 3 Apply the accepted measurement to covariance and accumulate δx .
- 4 Inject the accumulated error once; apply the rotation reset Jacobian.

Why gate against the prior?

It makes each landmark decision independent of arbitrary observation order. The covariance may shrink during the sweep, but the inlier decision should describe the same frame-entry hypothesis.

Appendix: landmark augmentation carries correlation forward

- Triangulation produces a world landmark and a 3D covariance propagated from independent pixel noise.
- The new landmark is correlated with robot state because the robot pose participates in the world-frame transform.
- The joint covariance is augmented with robot-landmark and landmark-landmark blocks; ignoring those correlations produces an over-confident map.

Tested properties

Symmetric PSD covariance, viewing-ray principal axis, Monte Carlo agreement within 3%, and degeneracy rejection for small disparity.

Appendix: the trace makes every presentation plot reproducible

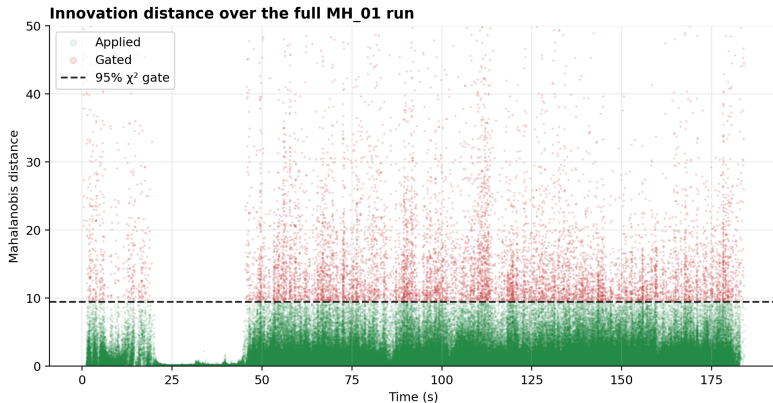
File	Contents	
imu_trace.csv	IMU timestamp, state, 15x15 robot covariance, interpolated truth	CSV
camera_trace.csv	Camera-time prior/posterior, update counts, track count, truth state/bias	
observation_trace.csv	Feature ID, observed/predicted pixels, innovation, Mahalanobis distance, outcome	
metadata.json	Dataset, code revision, initialization, noise, landmark budget, covariance layout	

is intentionally ignored by Git because one full run is hundreds of MB; generation code and metadata make the experiment reproducible without checking in bulk data.

Appendix: real-run metrics and controlled metrics answer different questions

Metric	Meaning	Correct interpretation
ATE	Global raw-frame position RMSE	No alignment; exposes accumulated drift
RPE	Local drift over a fixed interval	Separates local pose quality from global drift
NEES	Error relative to reported co-variance	Ensemble mean needs chi-square bounds
Innovation	Pixel residual relative to S	Gate / calibration diagnostic, not match quality alone

Appendix: full MH_01 trace adds a time-resolved innovation view



Values are clipped visually at 50 to keep the gate region legible.

Appendix: profiling changed the optimization plan

Stage	Before fix	After fix
Frontend	110.674 ms/frame	36.160 ms/frame
Mean NEES (15-dof)	217,927	10,357

The largest cost was feature

acquisition, not pyramidal KLT. A premature return in the coarse-to-fine KLT routine caused tracks to die early; correcting it improved both frontend cost and estimator behavior.

Appendix: questions to expect

Why not a smoother? Fixed compute budget, algorithmic clarity, and a real-time deployment goal. The cost is accuracy relative to batch optimization.

Why not MSCKF? This project deliberately keeps metric XYZ landmarks and joint covariance visible; the $O(n^2)$ state cost is understood and documented.

Is 1.9 us per IMU step a Jetson number? No. It is a RelWithDebInfo development-machine number; target hardware validation remains future work.

Is the filter consistent? Propagation is consistent in controlled Monte Carlo. The current camera-update path is measured as over-confident; that is a known, isolated limitation.

The most defensible claim is not “I implemented an EKF.”

I built an estimator whose assumptions,
numerical behavior, and failures are visible.